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Teil 1: Grundlagen,
Englische Übersetzung von DIN 743-1:2012-12

Calcul de la capacité des abres et axes –
Partie 1: Base,
Traduction anglaise de DIN 743-1:2012-12

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Foreword

This standard has *been prepared* by Working Committee NA 060-34-32 AA *Wellen- und Welle-Nabe-Verbindungen* of Section *Antriebstechnik* of the *Normenausschuss Maschinenbau* (Mechanical Engineering Standards Committee) in DIN.

DIN 743, *Calculation of load capacity of shafts and axles* comprises:

— *Part 1: General*

— *Part 2: Theoretical stress concentration factors and fatigue notch factors*

— *Part 3: Strength of materials*

— *Part 4: Fatigue limit, endurance limit — Equivalently damaging continuous stress*

— *Supplement 1: Examples to part 1 to 3*

— *Supplement 2: Examples to part 4*

Amendments

This standard differs from DIN 743-1:2000-10 as follows:

a) Clause 5 has been revised:

   — for the verification of structural strength, a distinction has been made between proof of avoidance of permanent deformation and proof of avoidance of incipient cracks in hard surfaces;

   — additions have been made to the determination of the component incipient crack limit;

b) editorial changes have been made.

Previous editions

DIN 743-1: 2000-10

Introduction

A large number of machine failures can be traced back to damage to axles and shafts. The most common cause is fatigue failure (endurance failure). Beside optimal design, the calculation of the factor of safety against fatigue failure and damage due to maximum loading (permanent deformation, incipient cracking) is a necessary measure.

This standard contains the fundamental equations and the systematic procedure for verifying the load capacity of axles and shafts. This proof is furnished by determining a (calculated) factor of safety (against fatigue failure and damage due to maximum loading). This safety factor takes into account both uncertainties in the calculation method used and the assumed loading data and also the importance of the system itself as well as consequential damage.

1 Scope

This standard applies to the calculation and verification of the safety of axles and shafts against:

— **fatigue failure** (endurance failure) if the fatigue strength is exceeded;

— **permanent deformation** (or incipient cracking or overload breakage).

The calculation of the factor of safety against fatigue failure is based on constant equivalent damaging stress amplitudes. These result from the agreed loadings or are to be determined by appropriate failure theories. For the calculation of the factor of safety against permanent deformation, the maximum possible stress is used. This is obtained by using the agreed or determined maximum load (see [1]).

The scope of this standard is limited to **steel**. Welded components are to be considered separately, and this standard does not apply to them.

To calculate the fatigue strength it is assumed that the amplitudes of the individual types of loading (tension/compression, bending, torsion) occur at the same point in time.

Residual stresses are not explicitly considered in the calculation. In the case of hard surfaces, they are globally taken into account by the sensitivity factor n and the influence factor for surface conditioning K_V.

The change in fatigue strength due to residual stress caused by local deformation has not yet been sufficiently investigated, so that this influence cannot yet be calculated. For boundary cases (components the failure of which results in heavy consequential damage), experimental studies or, alternatively, monitoring within the scope of inspections, may be required, in particular when overloading causes residual tensile stresses. As to the current situation regarding ductile materials, it has not yet been established that local plastic deformation and thus, residual stresses, have an essential influence on fatigue strength.

Deviations from the standard are permissible if, through theory or experimentally, the reliability of the design has been proven.

If in the following, reference is only made to “shafts”, inclusion of “axles” shall also be implicitly understood.

Limits of application: The capacity data apply for:

— tension/compression, bending, torsion as individual stresses and combined as alternating or pulsating stresses; without dominating shear stress; without buckling (due to compressive stress);

— a temperature range of $-40\ °C \leq \vartheta \leq 150\ °C$;

— rotational bending and flat bending without differentiation between the two (the ultimate strength values in DIN 743-3 for bending stress were obtained by rotational bending fatigue tests; the data are thus based on the more unfavourable case);

— corrosion-free surroundings (air, acid-free oil).

The ultimate strength values for safety against fatigue failure have been obtained for the ultimate number of cycles $N_G = 10^7$ and are set up as fatigue strength values. Recent tests show that in the case of high cycle fatigue, the fatigue strength can continue to taper off. This is attributed to the effect of micro inclusions. The correlations have not yet been sufficiently investigated. It is recommended that higher safety factors be agreed for components the failure of which leads to severe consequential damage.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

DIN 743-2, *Calculation of load capacity of shafts and axles — Part 2: Theoretical stress concentration factors and fatigue notch factors*

DIN 743-3, *Calculation of load capacity of shafts and axles — Part 3: Strength of materials*

DIN EN ISO 18265, *Metallic materials — Conversion of hardness values*

3 Symbols, designations and units

Symbol	Designation	Unit
d	Component diameter at notch cross section	mm
d_B	Reference diameter	mm
d_{eff}	Diameter relevant for heat treatment	mm
d_i	Inside diameter (bore diameter)	mm
n	Sensitivity factor	
r	Notch radius	mm
F	Force	N
G	Stress gradient	N/mm^3
G'	Relative stress gradient	mm^{-1}
H	Auxiliary factor (for negative mean stress)	N^2/mm^4
$K_{F\sigma}$, $K_{F\tau}$	Influence factor for surface roughness	
K_V	Influence factor for surface conditioning	
$K_1(d_{eff})$	Technological size influence factor	
$K_2(d)$	Geometrical size influence factor (for the unnotched, polished round bar)	
K_{2F}	Static support factor	
$K_3(d)$	Geometrical size influence factor (for the fatigue notch factor)	
$K_{\sigma,\tau}$	Total influence factor (bending or tension/compression and torsion)	
M_b	Bending moment	Nm
M_t	Torsional moment	Nm

Symbol	Designation	Unit
R_Z	Average peak-to-valley height (roughness)	µm
S	Calculated factor of safety	
S_{min}	Required minimum factor of safety	
α_σ, α_τ	Stress concentration factor	
β_σ, β_τ	Fatigue notch factor	
γ_F	Increase factor for yield point	
σ_a, τ_{ta}	Stress amplitude (principal or nominal stress)	N/mm^2
σ_B, (R_m)	Tensile strength	N/mm^2
σ_{bF}	Bending yield point	N/mm^2
σ_m, τ_m	Mean stress (principal or nominal stress)	N/mm^2
σ_{max}	Max. nominal stress (general)	N/mm^2
σ_S, $\sigma_{0,2}$, (R_e, $R_{p0,2}$)	Yield strength	N/mm^2
σ_o, τ_o	Maximum stress (principal or nominal stress)	N/mm^2
σ_u, τ_u	Minimum stress (principal or nominal stress)	N/mm^2
$\sigma_{zd,bADK}$, τ_{tADK}	Stress amplitude of component fatigue strength for defined mean stress	N/mm^2
$\sigma_{zd,bFK}$, τ_{tFK}	Component yield point	N/mm^2
$\sigma_{zd,bODK}$, τ_{tODK}	Maximum stress for component fatigue strength for defined mean stress	N/mm^2
$\sigma_{zd,bUDK}$, τ_{tUDK}	Minimum stress for component fatigue strength for defined mean stress	N/mm^2
$\sigma_{zd,bW}$, τ_{tW}	Material fatigue limit under reversed stress for reference diameter d_B	N/mm^2
$\sigma_{zd,\ bWK}$, τ_{tWK}	Component fatigue limit under reversed stress	N/mm^2
τ_{tF}	Torsion yield point	N/mm^2
$\psi_{\sigma K}$, $\psi_{\tau K}$	Influence factor for mean stress sensitivity	

Subscipts

A	Endurable amplitude
a	Existing amplitude
b	Bending
bW	Alternating bending
D	Fatigue strength
K	Notched component
max	Maximum
t, τ	Torsion
σ	Bending, tension/compression
v	von Mises stress (combined effective stresses)
W	Alternating
zd	Tension/compression

4 Proof of avoidance of fatigue failure

4.1 Factor of safety

The calculated factor of safety, S, shall be equal to or greater than the minimum safety factor S_{min}:

$$S \geq S_{min} \tag{1}$$

The principles of the calculation method itself require a minimum safety factor S_{min} = 1,2.

Uncertainties with the assumed load, possible consequential damage, etc. require higher safety factors. These shall be agreed upon or otherwise determined.

The calculated safety factor is obtained by taking into account bending, tension/compression and torsion, assuming phase balance.

$$S = \frac{1}{\sqrt{\left(\frac{\sigma_{zda}}{\sigma_{zdADK}} + \frac{\sigma_{ba}}{\sigma_{bADK}}\right)^2 + \left(\frac{\tau_{ta}}{\tau_{tADK}}\right)^2}} \tag{2}$$

If only bending or torsion is present, then

for bending: $$S = \frac{\sigma_{bADK}}{\sigma_{ba}} \tag{3}$$

for torsion: $$S = \frac{\tau_{tADK}}{\tau_{ta}} \tag{4}$$

Equation (3) is likewise valid for tension/compression by replacing σ_{ba} with σ_{zda} and σ_{bADK} with σ_{zdADK}. Special calculations are required where dominant shear stress occurs.[1)]

In the above,

σ_{zda}, σ_{ba}, τ_{ta} are the amplitudes of the active stresses resulting from the external load in the form of tension/compression, bending, and torsion (according to Table 1) and

σ_{zdADK}, σ_{bADK}, τ_{tADK} are the endurable amplitudes (strength for tension/compression, bending and torsion) according to 4.3.

4.2 Working stresses

The amplitudes and mean values of the working stresses are calculated using the equations in Table 1.

Table 1 — Determination of working stresses

Type of stress	Working stress		Cross-sectional area and/or section modulus
	Amplitude	Mean value	
Tension/compression	$\sigma_{zda} = \frac{F_{zda}}{A}$	$\sigma_{zdm} = \frac{F_{zdm}}{A}$	$A = \frac{\pi}{4} \cdot (d^2 - d_i^2)$
Bending	$\sigma_{ba} = \frac{M_{ba}}{W_b}$	$\sigma_{bm} = \frac{M_{bm}}{W_b}$	$W_b = \frac{\pi}{32} \cdot \frac{(d^4 - d_i^4)}{d}$
Torsion	$\tau_{ta} = \frac{M_{ta}}{W_t}$	$\tau_{tm} = \frac{M_{tm}}{W_t}$	$W_t = \frac{\pi}{16} \cdot \frac{(d^4 - d_i^4)}{d}$
NOTE In the compression zone, σ_{zdm} and σ_{bm} are negative; see Figures A.1, A.2 (Annex A).			

In the above,

F_{zda}, M_{ba}, M_{ta} are the amplitudes of the working external load and/or the amplitudes of load (rotational bending) acting on the material fibre (e.g. on account of the shaft rotation);

F_{zdm}, M_{bm}, M_{tm} are the mean values of the working external load.

4.3 Fatigue strength value

The fatigue strength $\sigma_{zd,bADK}$, (τ_{tADK}) of the component is to be calculated from the strength of the polished test bar. It is given as nominal stress and represents the maximum endurable amplitude of the component for the load case in question. The following is taken into account:

— the heat treating quality and/or hardenability: if not directly known, for example from hardness measurements taken at the cross section in question, approximately dependent on the component diameter (technological size influence factor $K_1(d_{eff})$);

— the transition from the bending vibration strength to tension/compression vibration strength with increasing diameter as a result of the decrease of the stress gradient (geometrical size influence factor $K_2(d)$);

1) Among other things, the changes in the stress concentration factors and in the location of the primary stress shall be taken into account.

— the shape of the component, especially notches (fatigue notch factor $\beta_\sigma(d)$, $\beta_\tau(d)$),

— surface roughness (roughness factor $K_{F\sigma}$ and/or $K_{F\tau}$);

— the influence of surface hardening and compressive residual stresses acting on the surface (factor for surface conditioning K_V);

— the influence of the mean stress on the endurable stress amplitude (strength), (factor for mean stress sensitivity $\psi_{\sigma K}$ and/or $\psi_{\tau K}$).

It is desirable to start with the fatigue limit under reversed stresses existing on the component in question and at the point to be calculated, e.g. calculated from the hardness measured at that point. If the required conditions do not exist, $\sigma_{zdW}(d)$, $\sigma_{bW}(d)$ and $\tau_{tW}(d)$ can be approximately obtained from $\sigma_{zdW}(d_B)$, $\sigma_{bW}(d_B)$, $\tau_{tW}(d_B)$ for the test bar diameter d_B (reference diameter) and a size factor $K_1(d_{eff})$, (e.g. $\sigma_{bW}(d_B) \cdot K_1(d_{eff}) \approx \sigma_{bW}(d)$).

The component fatigue strength, taking into account the above influences, is calculated using Equations (10) to (14) and (15) to (19) with the aid of Equations (5) to (7) (component fatigue limit under reversed stress) and (8), (9). The *fatigue limit under reversed stress of the (notched) component* is:

$$\sigma_{zdWK} = \frac{\sigma_{zdW}(d_B) \cdot K_1(d_{eff})}{K_\sigma} \quad 2) \tag{5}$$

$$\sigma_{bWK} = \frac{\sigma_{bW}(d_B) \cdot K_1(d_{eff})}{K_\sigma} \tag{6}$$

$$\tau_{tWK} = \frac{\tau_{tW}(d_B) \cdot K_1(d_{eff})}{K_\tau} \tag{7}$$

where

$K_1(d_{eff})$ is the technological size influence factor (hardenability, heat treating quality according to DIN 743-2) for tensile strength;

$\sigma_{zdW}(d_B)$, $\sigma_{bW}(d_B)$, $\tau_{tW}(d_B)$ are the fatigue limits under reversed stress of the polished test bar for reference diameter d_B according to DIN 743-3.

The total influence factor K_σ for tension/compression and bending

$$K_\sigma = \left(\frac{\beta_\sigma}{K_2(d)} + \frac{1}{K_{F\sigma}} - 1 \right) \cdot \frac{1}{K_V} \tag{8}$$

and K_τ for torsion

$$K_\tau = \left(\frac{\beta_\tau}{K_2(d)} + \frac{1}{K_{F\tau}} - 1 \right) \cdot \frac{1}{K_V} \tag{9}$$

is to be determined by the following quantities according to DIN 743-2:

2) $K_1(d_{eff})$ is to be determined from the tensile strength (see DIN 743-2, Figure 13).

β_σ is the fatigue notch factor for tension/compression and bending (in the case of torsion β_τ);

$K_2(d)$ is the geometrical size influence factor (drop of σ_{bW} relative to σ_{zdW} with increasing diameter; torsion likewise);

$K_{F\sigma}$ is the influence factor for surface roughness for tension/compression and bending (with torsion $K_{F\tau}$);

K_V is the influence factor for surface conditioning (e.g. by shot peening or surface hardening).

The fatigue strength is to be calculated as a function of the relation between changes in the decisive stresses and an increase in the loads. Here, a distinction is made between two load cases. (In case of doubt, if $\sigma_{zdm} + \sigma_{bm} \geq 0$, calculation can be made taking load case 2 as a basis.).

Case 1 (σ_{mv} = constant and/or τ_{mv} = constant, see Figure A.4):

Case 1 applies if with a change in the operating load the stress amplitude changes and the mean stress remains constant.

Under the condition $\sigma_{mv} \leq \frac{\sigma_{zd,bFK} - \sigma_{zd,bWK}}{1 - \psi_{zd,b\sigma K}}$ and/or $\tau_{mv} \leq \frac{\tau_{tFK} - \tau_{tWK}}{1 - \psi_{\tau K}}$

the endurable amplitude for σ_{mv} = constant (τ_{mv} = constant) is:

$$\sigma_{zdADK} = \sigma_{zdWK} - \psi_{zd\sigma K} \cdot \sigma_{mv} \quad (10)$$

$$\sigma_{bADK} = \sigma_{bWK} - \psi_{b\sigma K} \cdot \sigma_{mv} \quad (11)$$

$$\tau_{tADK} = \tau_{tWK} - \psi_{\tau K} \cdot \tau_{mv} \quad (12)$$

If the condition mentioned for Equations (10) to (12) is not met, the *endurable amplitude* for σ_{mv} = constant (τ_{mv} = constant) is:

$$\sigma_{zd,bADK} = \sigma_{zd,bFK} - \sigma_{mv} \quad (13)$$

$$\tau_{tADK} = \tau_{tFK} - \tau_{mv} \quad (14)$$

If $\sigma_{zdm} + \sigma_{bm} < 0$, in place of σ_{mv} (according to Equation (23)) the calculation shall be made with

$$\sigma_{mv} = \frac{H}{|H|} \cdot \sqrt{|H|}\,; \qquad H = \frac{(\sigma_{bm} + \sigma_{zdm})^3}{|\sigma_{bm} + \sigma_{zdm}|} + 3 \cdot \tau_{tm}{}^2$$

and with τ_{mv} according to Equation (24).

If $\sigma_{mv} < 0$, $\tau_{mv} = 0$ shall be used. Equations (10) to (12) apply for direct stresses, if $\sigma_{mv} \geq \sigma_{mvgrenzF1}$ (see Figure A.5, Annex A). If this condition is not met, Table A.1 (Annex A) shall be used.

Case 2 ($\sigma_{mv}/\sigma_{zd,ba}$ = constant and/or τ_{mv}/τ_{ta} = constant, see Figure A.4):

Case 2 applies, if with a change in the operating load the relation between amplitude of stress and mean stress remains constant. Under the condition

$$\frac{\sigma_{mv}}{\sigma_{zd,ba}} \leq \frac{\sigma_{zd,bFK} - \sigma_{zd,bWK}}{\sigma_{zd,bWK} - \sigma_{zd,bFK} \cdot \psi_{zd,b\sigma K}} \quad \text{and/or} \quad \frac{\tau_{mv}}{\tau_{ta}} \leq \frac{\tau_{tFK} - \tau_{tWK}}{\tau_{tWK} - \tau_{tFK} \cdot \psi_{tK}}$$

the endurable amplitude for $\sigma_{mv}/\sigma_{zd,ba}$ = constant (τ_{mv}/τ_{ta} = constant) is:

$$\sigma_{zdADK} = \frac{\sigma_{zdWK}}{1 + \psi_{zd\sigma K} \cdot \frac{\sigma_{mv}}{\sigma_{zda}}} \tag{15}$$

$$\sigma_{bADK} = \frac{\sigma_{bWK}}{1 + \psi_{b\sigma K} \cdot \frac{\sigma_{mv}}{\sigma_{ba}}} \tag{16}$$

$$\tau_{tADK} = \frac{\tau_{tWK}}{1 + \psi_{\tau K} \cdot \frac{\tau_{mv}}{\tau_{ta}}} \tag{17}$$

If the condition mentioned for Equations (15) to (17) is not fulfilled, the endurable amplitude for $\sigma_{mv}/\sigma_{zd,ba}$ = constant (τ_{mv}/τ_{ta} = constant) is:

$$\sigma_{zd,bADK} = \frac{\sigma_{zd,bFK}}{1 + \frac{\sigma_{mv}}{\sigma_{zd,ba}}} \tag{18}$$

$$\tau_{tADK} = \frac{\tau_{tFK}}{1 + \frac{\tau_{mv}}{\tau_{ta}}} \tag{19}$$

where

$\psi_{\sigma K}$, $\psi_{\tau K}$ are the influence factors for mean stress sensitivity for tension/compression, bending and torsion according to Equations (20) to (22);

σ_{zdWK}, σ_{bWK}, τ_{tWK} are according to Equations (5) to (7);

σ_{zda}, σ_{ba}, τ_{ta} are the stress amplitudes according to Table 1;

σ_{mv}, τ_{mv} are combined mean stresses (von Mises) according to Equations (23) and (24);

$\sigma_{zd,\ bFK}$, τ_{tFK} are the component yield points according to Equation (31) and/or (32).

If $\sigma_{zdm}+\sigma_{bm} < 0$, in place of σ_{mv} (according to Equation (23)) the calculation shall be made with

$$\sigma_{mv} = \frac{H}{|H|} \cdot \sqrt{|H|}\,; \qquad H = \frac{(\sigma_{bm} + \sigma_{zdm})^3}{|\sigma_{bm} + \sigma_{zdm}|} + 3 \cdot \tau_{tm}{}^2$$

and with τ_{mv} by Equation (24). If $\sigma_{mv} < 0$, $\tau_{mv} = 0$ shall be used.

The condition for the validity of Equations (15) to (17) for direct stresses is

$$\frac{\sigma_{mv}}{\sigma_{zd,ba}} > \left(\frac{\sigma_{mv}}{\sigma_{zd,ba}}\right)_{grenzF2}$$

If this condition is not fulfilled, Table A.1 and Figure A.5 (Annex A) shall be used.

The influence factors for mean stress sensitivity shall be calculated using Equations (20) to (22):

$$\psi_{zd\sigma K} = \frac{\sigma_{zdWK}}{2 \cdot K_1(d_{eff}) \cdot \sigma_B(d_B) - \sigma_{zdWK}} \tag{20}$$

$$\psi_{b\sigma K} = \frac{\sigma_{bWK}}{2 \cdot K_1(d_{eff}) \cdot \sigma_B(d_B) - \sigma_{bWK}} \tag{21}$$

$$\psi_{\tau K} = \frac{\tau_{tWK}}{2 \cdot K_1(d_{eff}) \cdot \sigma_B(d_B) - \tau_{tWK}} \tag{22}$$

where

$K_1(d_{eff})$ is the technological size influence factor (heat treating quality, hardenability) according to DIN 743-2 for tensile strength;

σ_B is the tensile strength for test bar diameter d_B.

The combined mean stresses (von Mises) shall be calculated using Equations (23) and (24):

$$\sigma_{mv} = \sqrt{(\sigma_{zdm} + \sigma_{bm})^2 + 3 \cdot \tau_{tm}{}^2} \tag{23}$$

$$\tau_{mv} = \frac{\sigma_{mv}}{\sqrt{3}} \tag{24}$$

5 Proof of avoidance of permanent deformation, incipient cracking or overload breakage under maximum load

5.1 Safety factor

The calculated safety factor S shall be equal to or greater than the minimum safety factor S_{min} ($S > S_{min}$; see explanations of Equation (1)). The principles of the calculation method require a minimum safety factor S_{min} = 1,2. Uncertainties in the estimation of the maximum load, possible consequential damage, etc. require higher safety factors. These shall be agreed upon or otherwise determined. On principle, it shall be established that permanent deformation and incipient cracks are avoided. If there is no risk of brittle fracture ($\sigma_B \leq 1\,300$ N/mm^2), incipient cracks and overload breakage do not generally occur on structural and quenched and tempered steels at maximum load within the usual area of application prior to permanent component deformation. In this case it is sufficient to prove that permanent deformation of the macro geometry is avoided.

Also on shafts with a hard surface (e.g. case hardened shafts), permanent component deformation can occur prior to an incipient cracking (mainly dependent on the stress concentration at the notch and the core hardness). Since the hardened case is not ductile it shall be proved that permanent deformation below the case **and** incipient cracking and/or overload breakage in the case are avoided.

If $\alpha \cdot \sigma_{max} > \sigma_{0,2} \rightarrow \sigma_B$, it shall be checked on tempering steels and high-tensile steels with σ_B > 1 300 N/mm^2 whether the ductility is sufficient to reduce the stress peak by plastic deformation.

An incipient crack does not yet occur with $\sigma_{max} \leq \sigma_{0,2}$ and a stress concentration factor of $\alpha \leq 10$ and at least 4 % local plastic ductility of the material. The local plastic ductility is greater than the elongation on fracture. This can serve as a rough guide if no specially determined values are available.

For $\sigma_{max} > \sigma_{0,2}$, stress calculations shall be made following highly sophisticated analysis methods (e.g. FEM, BEM) or by carrying out experimental tests to check the risk of incipient cracking.

5.1.1 Proof of avoidance of permanent deformation

Proof of avoidance of permanent deformation shall be furnished. It does not refer to the avoidance of local deformations (e.g. in the notch root), but to the avoidance of permanent deformations in larger areas of the component (unacceptable dimensional deviations, deviations exceeding the tolerance value). In the case of hard surfaces, the avoidance of permanent deformations below the hard surface shall be examined. For such areas it is assumed that the notch effect has faded. The calculated factor of safety against permanent deformation resulting from combined stresses (composed of tension/compression, bending, and torsion) shall be calculated with Equation (25) taking into account the scope of validity mentioned in 5.1. Compressive stresses shall be used in Equation (25) with a negative sign.

$$S = \frac{1}{\sqrt{\left(\frac{\sigma_{zdmax}}{\sigma_{zdFK}} + \frac{\sigma_{bmax}}{\sigma_{bFK}}\right)^2 + \left(\frac{\tau_{tmax}}{\tau_{tFK}}\right)^2}} \tag{25}$$

If only bending or torsion is present, then

for bending: $$S = \frac{\sigma_{bFK}}{\sigma_{bmax}} \tag{26}$$

for torsion: $$S = \frac{\tau_{tFK}}{\tau_{tmax}} \tag{27}$$

(Equation (26) is likewise valid for tension/compression by replacing σ_{bmax} with σ_{zdmax} and σ_{bFK} with σ_{zdFK}.)

In the above

σ_{zdFK}, σ_{bFK}, τ_{tFK} are the component yield points for tension/compression, bending and/or torsion (see 5.2)

σ_{zdmax}, σ_{bmax}, τ_{tmax} are the existing maximum nominal stresses as a result of the operating load. They are determined by means of Table 5, using the maximum occurring loads F_{zdmax}, M_{bmax} and M_{tmax}.

NOTE In the case of materials for highly stressed shafts with hard surfaces it is recommended to consider the actual hardness and/or strength progression into the core of the material and compare it with the slope of stress. If no knowledge is available here, calculations to prove the avoidance of plastic deformation below the surface can be made by approximation with the maximum nominal stress on the surface, and this maximum nominal stress can then be compared with the yield strength of the core.

5.1.2 Proof of avoidance of incipient cracking (and/or overload breakage) in hard surfaces

The calculated factor of safety against incipient cracking and/or overload breakage as a result of combined stresses composed of tension/compression, bending, and torsion shall be calculated using Equation (28) (direct stress theory):

$$S = \frac{1}{0{,}5 \cdot \left[\frac{\alpha_{\sigma zd} \cdot \sigma_{zdmax}}{\sigma_{zdBRand}} + \frac{\alpha_{\sigma b} \cdot \sigma_{bmax}}{\sigma_{bBRand}} + \sqrt{\left(\frac{\alpha_{\sigma zd} \cdot \sigma_{zdmax}}{\sigma_{zdBRand}} + \frac{\alpha_{\sigma b} \cdot \sigma_{bmax}}{\sigma_{bBRand}} \right)^2 + \left(\frac{2 \cdot \alpha_{\tau} \cdot \tau_{tmax}}{\tau_{tBRand}} \right)^2} \right]} \tag{28}$$

Here, the local stresses are decisive.

If bending or torsion only is present, then

for bending: $$S = \frac{\sigma_{bBRand}}{\sigma_{bmax} \cdot \alpha_{\sigma b}} \tag{29}$$

for torsion: $$S = \frac{\tau_{tBRand}}{\tau_{tmax} \cdot \alpha_{\tau}} \tag{30}$$

(Equation (29) is likewise valid for tension/compression by replacing σ_{bmax} with σ_{zdmax} and $\sigma_{bB\,Rand}$ with $\sigma_{zdB\,Rand}$.)

where

$\alpha_{\sigma zd,b}$, α_{τ} are the stress concentration factors for tension/compression, bending and/or torsion

$\sigma_{zdBRand}$, σ_{bBRand}, τ_{tBRand} are the breaking points in the hard surface in the case of tension/compression, bending and/or torsion (see 5.3)

If the stress concentration factors or local stresses are not known, the fatigue notch factors can be used by approximation.

5.2 Component yield point

The component yield point is taken as the basis of the calculation of the factor of safety against permanent deformation. The starting point is the yield strength $\sigma_S(d)$ at the cross section of the component in question. If this is not known, $\sigma_S(d)$ can be determined by approximation from the yield strength valid for the test bar diameter d_B (reference diameter) and a size factor $K_1(d_{eff})$ (i.e. $\sigma_S(d) = \sigma_S(d_B) \cdot K_1(d_{eff})$). When applying this method, the component yield point shall be calculated using Equation (31) and/or (32).

$$\sigma_{zd,bFK} = K_1(d_{eff}) \cdot K_{2F} \cdot \gamma_F \cdot \sigma_S(d_B) \tag{31}$$

$$\tau_{tFK} = K_1(d_{eff}) \cdot K_{2F} \cdot \gamma_F \cdot \sigma_S(d_B) / \sqrt{3} \tag{32}$$

In the above

$K_1(d_{eff})$ is the technological size influence factor (heat treating quality, hardenability) according to DIN 743-2 for the yield strength;

K_{2F} is the static support factor according to Table 3 as a result of local plastic deformation on the surface. On hard surfaces, K_{2F} = 1 (for the calculation to be made for the value below the surface);

γ_F is the factor increasing the yield point as a result of the multi-axial stress condition at the circumferential notch and local hardness increase according to Table 2. In the case of hard surfaces or no circumferential notch, γ_F = 1;

$\sigma_S(d_B)$ is the yield strength for the reference diameter d_B according to DIN 743-3; in the case of hard surfaces, the values for the core apply.

Table 2 — Increase factor for yield point γ_F at circumferential notch (α_σ and/or β_σ according to DIN 743-2) and materials without hard surface

Type of stress	α_σ or β_σ	γ_F
Tension/compression or bending	up to 1,5	1,00
	1,5 to 2,0	1,05
	2,0 to 3,0	1,10
	above 3,0	1,15
Torsion	Any	1
NOTE As a result of the multi-axial stress condition (e.g. also at circumferential notches), the yield point of the component is increased by γ_F, but the risk of low-ductility fractures increases.		

Table 3 — Static support factor K_{2F} for materials without hard surface

Type of stress	K_{2F}	
	Solid shaft	Hollow shaft
Tension/compression	1,0	1,0
Bending	1,2	1,1
Torsion	1,2	1,0

5.3 Component's incipient crack limit

Proof is required for hard surfaces and tempering steels with σ_B > 1 300 N/mm^2, if the local ductility is below 4 %. The incipient crack limits of the component shall be calculated on the basis of the resistances to breaking of the hard surfaces using Equation (33) and/or (34) (for brittle materials).

$$\sigma_{zd,bBRand} = \sigma_{BRand} \quad (33)$$

$$\tau_{tBRand} = \sigma_{BRand} \quad (34)$$

The starting point is the surface hardness on the real component. With a hardness of 700...800 HV (e.g. case hardness) calculations can use a tensile strength in the surface of σ_{BRand} = 2 300 N/mm^2, neglecting the residual stresses. If the surface has been treated or hardened in another way, the tensile strength in the surface can be approximately determined according to DIN EN ISO 18265:2004 (Table A.1). Tempering steels with nitrided or nitrocarburized surfaces do have tensile strengths of σ_{BRand} > 1 700 N/mm^2 in the surface.

NOTE σ_{BRand} is increased by the compressive residual stresses which generally exist in the hard surface.

5.4 Working stresses (maximum stresses)

The working stresses are to be calculated according to Table 4.

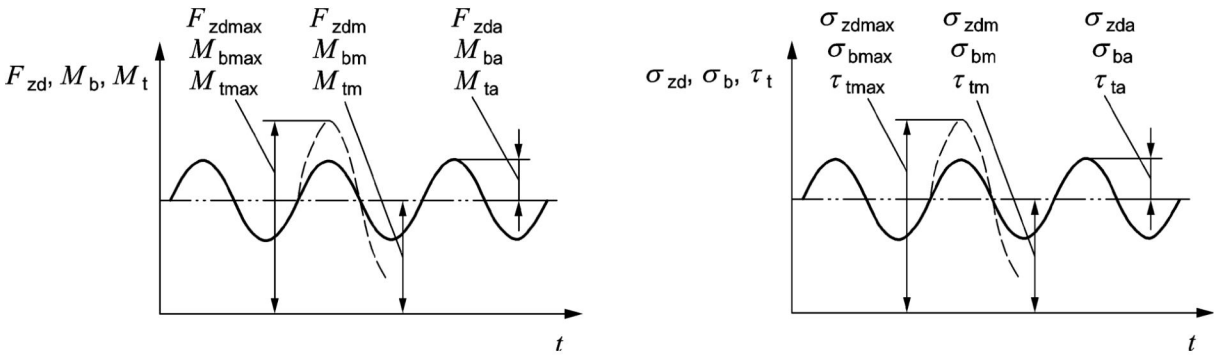
Table 4 — Determination of maximum stresses (maximum nominal stresses)

Type of stress	Working stress[3)]	Cross-sectional area and/or section modulus
Tension/compression	$\sigma_{zdmax} = \frac{F_{zdmax}}{A}$	$A = \frac{\pi}{4} \cdot (d^2 - d_i^2)$
Bending	$\sigma_{bmax} = \frac{M_{bmax}}{W_b}$	$W_b = \frac{\pi}{32} \cdot \frac{(d^4 - d_i^4)}{d}$
Torsion	$\tau_{tmax} = \frac{M_{tmax}}{W_t}$	$W_t = \frac{\pi}{16} \cdot \frac{(d^4 - d_i^4)}{d}$

3) See Figure A.1.

Annex A
(informative)

Explanations of the variation of load and/or stress, cross-sectional areas and the reading of σ_{ADK} from the Smith diagram



t = time

Figure A.1 — Variation of applied load with time (F_{zd}, M_b, M_t) and stress (σ_{zd}, σ_b, τ_t)

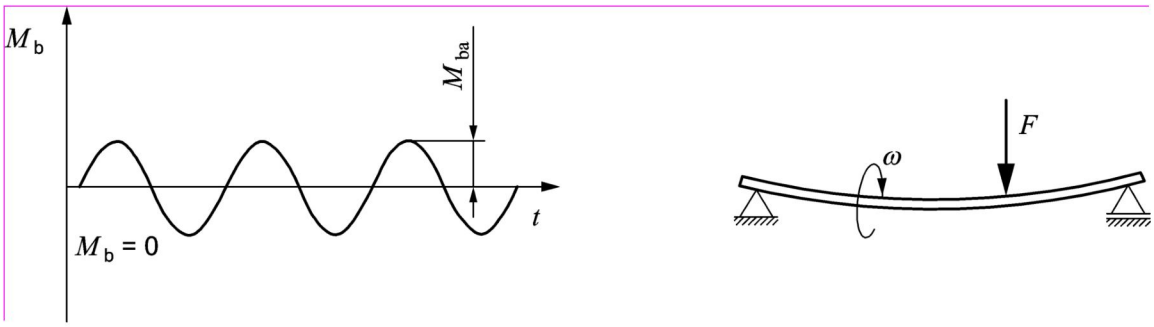


Figure A.2 — Origination of amplitude of bending moment M_b as a result of shaft rotation (rotational bending); force F with constant direction, shaft rotating $(\omega = 2 \cdot \pi \cdot n > 0)$

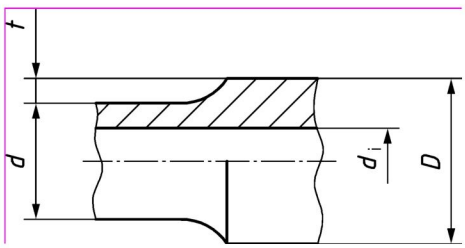
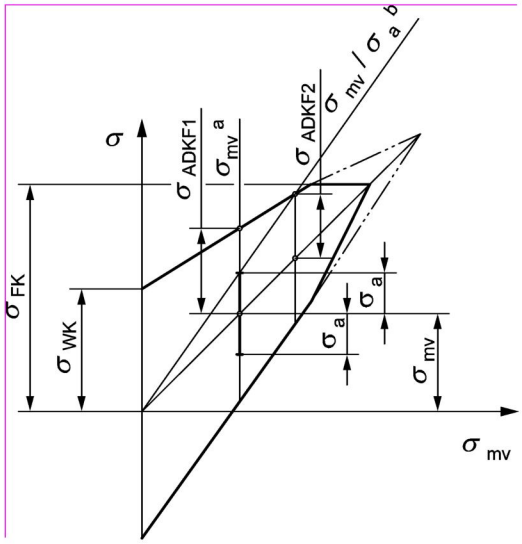


Figure A.3 — Cross section parameters

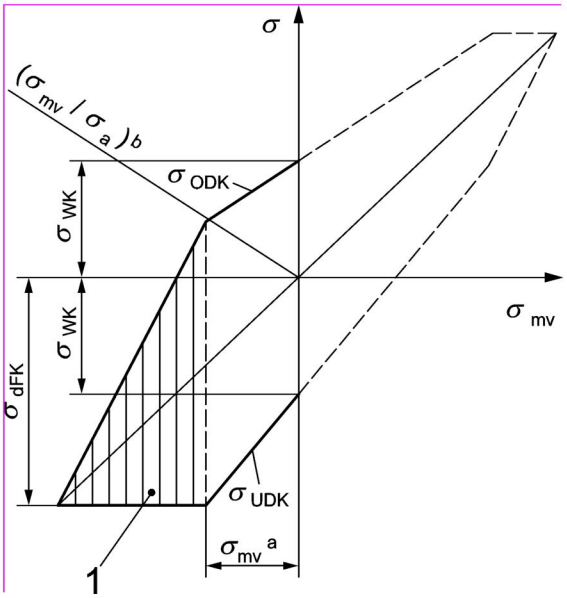


Key

F1 = Load case 1
F2 = Load case 2

NOTE The marked area in the Smith diagram is defined by the intersection point of the line σ_{UDK} extended into the compression zone with the line, given by the pressure yield point $\sigma_{dFK,}$ parallel to the abscissa. The boundary lines in the fatigue strength diagram (Smith diagram) σ_{ODK} and σ_{UDK} represent an approximation to the actual behaviour.

Figure A.4 — Load cases, represented in the fatigue strength diagram (Smith diagram)



Key

F1 = Load case 1
F2 = Load case 2

NOTE The marked area in the Smith diagram is defined by the intersection point of the line σ_{UDK} extended into the compression zone with the line, given by the pressure yield point $\sigma_{dFK,}$ parallel to the abscissa.

Figure A.5 — Fatigue strength diagram with extension of the compression zone (component pressure yield point σ_{dFK})

Table A1 — σ_{ADK} in the marked compression zone for load case 1 with $\sigma_{mv} < \sigma_{mv\ grenz\ F1}$ and/or for load case 2 with $\sigma_{mv}/\sigma_a < (\sigma_{mv}/\sigma_a)_{grenz\ F2}$ (in the compression zone $\sigma_{mv} < 0$ or $\sigma_{mv}/\sigma_a < 0$ applies)

Case	$\sigma_{mv\ grenz\ F1}$, $(\sigma_{mv}/\sigma_a)_{grenz\ F2}$	σ_{ADK}
Case 1(σ_{mv} = constant)	$\sigma_{mv\ grenz\ F1} = (\sigma_{zd,bWK} - \sigma_{dFK}) \cdot (1 - \frac{\sigma_{zd,bWK}}{2 \cdot \sigma_B(d)})$	$\sigma_{ADK} = \sigma_{mv} + \sigma_{dFK}$
Case 2 (σ_{mv}/σ_a = constant)	$\left(\frac{\sigma_{mv}}{\sigma_{zd,ba}}\right)_{grenz\ F2} = \frac{\sigma_{zd,bWK} - \sigma_{dFK}}{\psi_{zd,b\sigma K} \cdot \sigma_{dFK} + \sigma_{zd,bWK}}$	$\sigma_{ADK} = \frac{\sigma_{dFK} \cdot \sigma_{zd,ba}}{\sigma_{zd,ba} - \sigma_{mv}}$
If no other experience or no experimental data are available, $\sigma_{zFK}(d)$ can be used for σ_{dFK}. $\sigma_{zFK}(d) = \sigma_{zdFK}(d)$ according to Equation (31); σ_{dFK} and σ_{WK} are given positive values in the equations: $\sigma_{dFK,}\ \sigma_{WK} > 0$		

Annex B
(normative)

Schema of the safety factor calculation

B.1 General schema

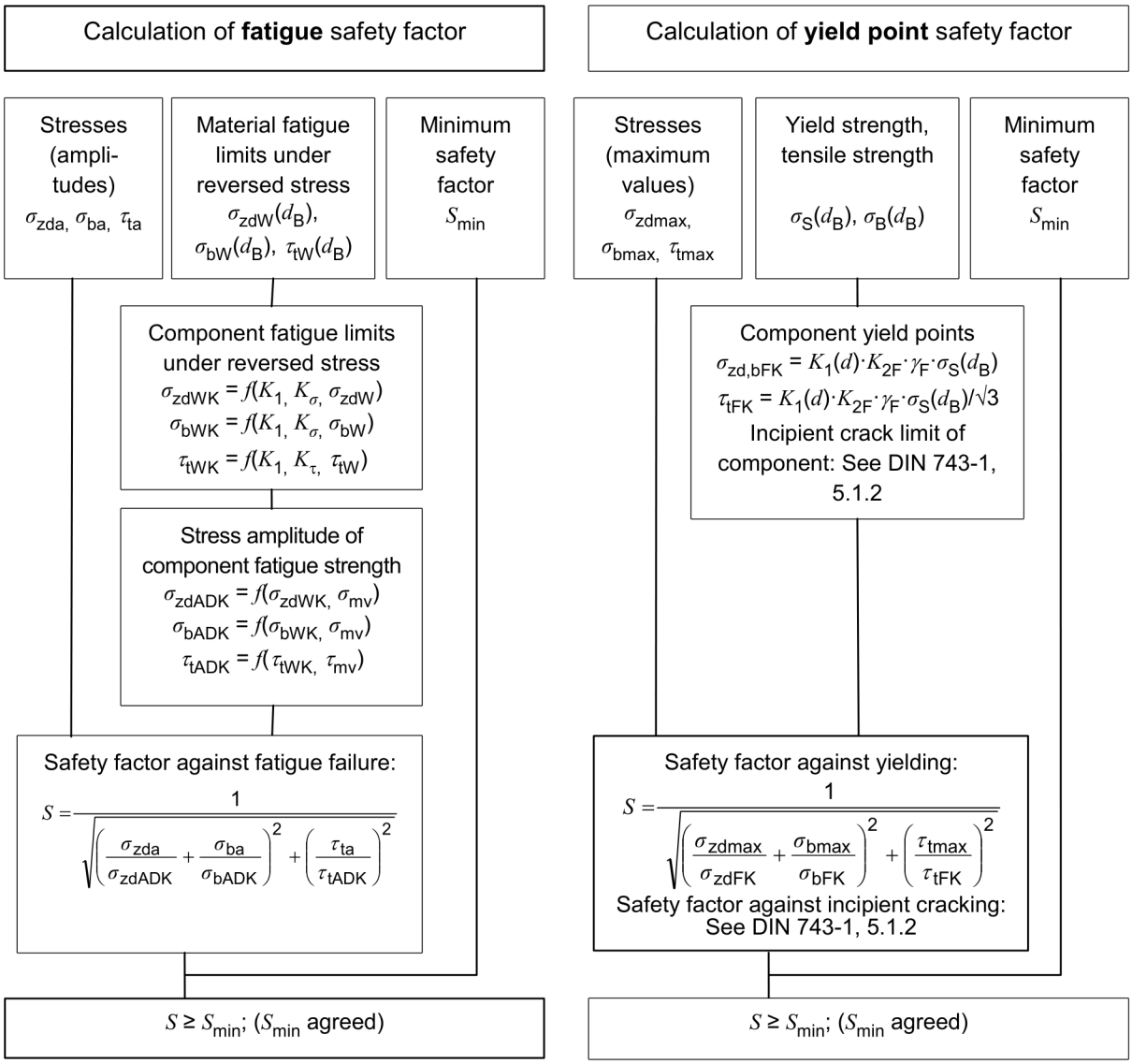


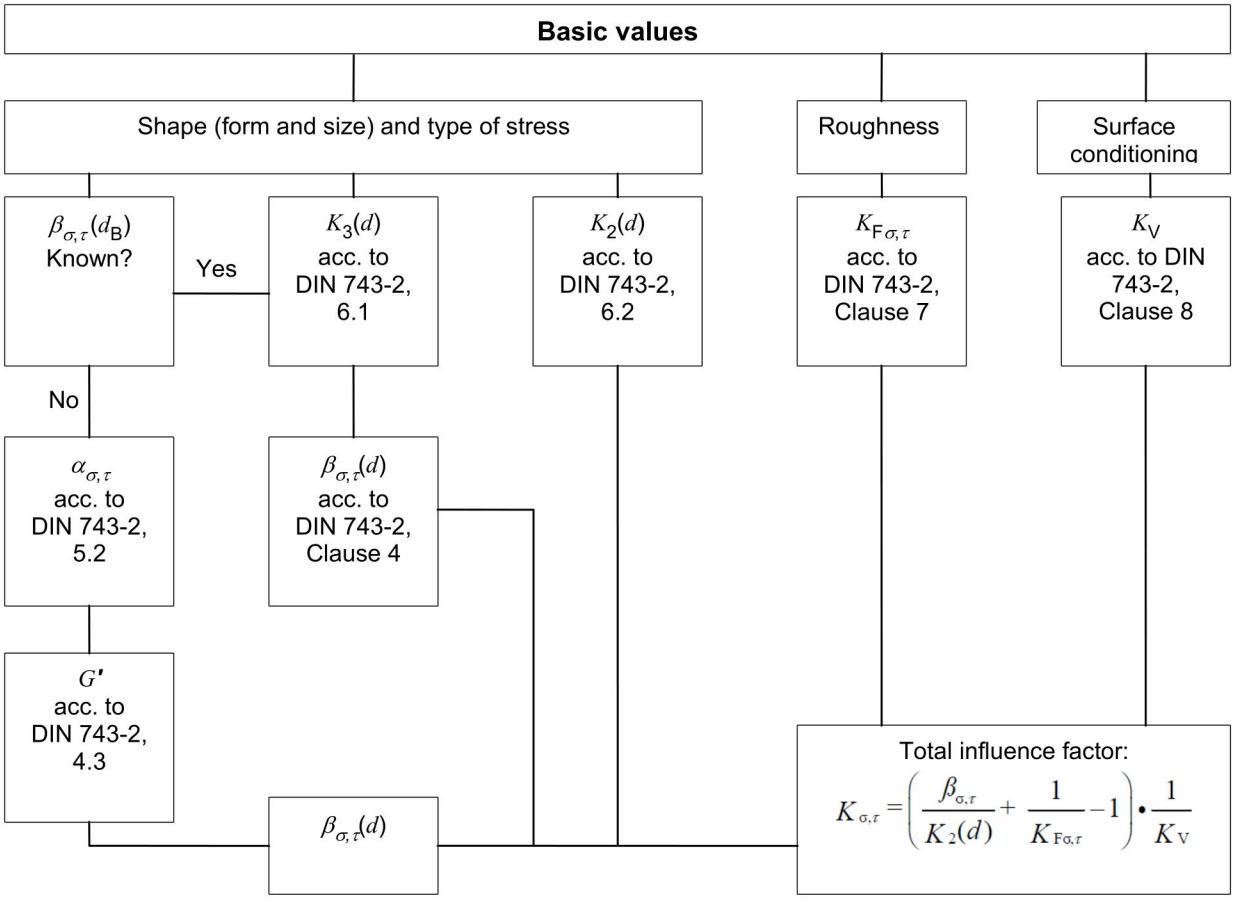
Figure B.1 *(continued)*

Key

σ_{WK}, τ_{WK}	Component fatigue limits under reversed stress according to DIN 743-3
σ_{zd}, σ_{bADK}, τ_{tADK}	Stress amplitude of component fatigue strength according to Eq. (5)…(7) with $K_{\sigma,\tau}$ according to Figure B.2, DIN 743-1
σ_{zdFK}, σ_{bFK}, τ_{FK}	Component yield points according to Eq. (31) and (32) DIN 743-1
σ_{zda}, σ_{ba}, τ_{ta}	Existing stress amplitudes according to Table 1 DIN 743-1
σ_{zdmax}, σ_{bmax}, τ_{tmax}	Maximum stresses according to Table 4 DIN 743-1
$K_{\sigma,\tau}$	Total influence factor according to Figure B.2
$K_1(d_{eff})$	Technological size influence factor according to Eq. (10)...(14), Figure 13, DIN 743-2
$K_2(d)$	Geometrical size influence factor according to Eq. (15), (16), Figure 14, DIN 743-2
K_{2F}	Static support factor according to Table 3 DIN 743-1
γ_F	Increase factor for yield point according to Table 2 DIN 743-1
S, S_{min}	Calculated or minimum safety factor

Figure B.1 — Calculation procedure for safety factors

B.2 Total influence factor



Key

$\beta_{\sigma,\tau}$	Fatigue notch factor (for tension/compression, bending and/or torsion)
$\alpha_{\sigma,\tau}$	Stress concentration factor (for tension/compression, bending and/or torsion)
$K_2(d)$, $K_3(d)$	Geometrical size influence factors (tension/compression, bending and/or torsion) considering the drop of σ_{bw} relative to σ_{zdw} with increasing diameter ($K_2(d)$) or dependence of fatigue notch factor on diameter ($K_3(d)$)
d	Diameter of component
d_B	Reference diameter
G'	Relative stress gradient (for tension/compression, bending and/or torsion)
n	Sensitivity factor (for tension/compression, bending and/or torsion)
r	Notch radius
$K_{F\sigma,\tau}$	Influence factor for surface roughness (for tension/compression, bending and/or torsion)
K_V	Influence factor for surface conditioning

Figure B.2 — Calculation procedure for total influence factor $K_{\sigma,\tau}$

Bibliography

[1] FKM Guideline: *Analytical strength assessment of components in mechanical engineering*, *Forschungskuratorium Maschinenbau (FKM), VDMA-Verlag Frankfurt/Main 2003*